

CS3 Rubric – Case Study Reproduction

DS 4002 – Spring 2026 - Instructors: Javier Rasero; TA: Ethan Meidinger
Due: TBD

Submission format:

- Slide presentation of case study process and results to be given in class. The goal is about 7 total slides to support a 5 minute presentation to peers.
- 1-2 page reflection on the reproduction process and what you learned through completing reproduction of results outlined in case study.

General Description: Submit to presentation slides and reflection on Canvas. Presentations will be given in class after submission of these items.

Preparatory Assignments – None. You only need what you know today, the goal is to grow new skills with the case.

Why am I doing this? This case is your opportunity to learn about methods of using image data in data science using both a fully pretrained model version and a custom trained model version for person detection. The case study is a chance to follow and recreate a project from someone who used to sit in the seat you are in today. Case studies developed from work by previous UVA DS students should excite you because as you continue your data science education you can expect to reach the level of execution to organically create similar projects (and likely even more nuanced and interesting)! The deliverable you will create is the presentation with the results of your production and a reflection on how this case impacted your data science skills and dexterity.

What am I going to do? In assignment CS3 *Case Study Reproduction* you will review the case study provided to you made by a previous UVA DS undergraduate. Now in CS3 you have the opportunity to recreate the quantitative results and demonstrate your learnings qualitatively through written reflection and a low-stake presentation of the process. You will stream in and clean a data set, work with the Ultralytics YOLOv8 (You-Only-Look-Once) on both pretrained weights and custom trained weights, and use your person detections in the movie posters from the model to do quantitative analysis to see if there are trends between the variables (popularity score vs. persons detected). You will produce two items through the case study reproduction process. Deliverables include:

- One-two page reflection document - reflection on what you did and what you learned from the case to add to your skill repertoire.
- Slides - 7 slides outlining the quantitative process of reproduction, your learnings from the results, and what skills you gained.
- Presentation to Peers - after submission, you will give a 5 minute presentation of your slide to your peers in class.

All of this will be submitted electronically via Canvas. The presentation will be given in person during class time.

Tips for success:

- Follow the workflow in the repository carefully.
- Read resources that explain the quantitative methods of reproduction.

How will I know I have Succeeded? You will meet expectations on CS3 *Create Case Study Reproduction* when you follow the criteria in the rubric below.

Reproduction	● <u>Goal</u>: Demonstrate ability to reproduce the case and generate output results from contents in GitHub repository. ● Stream in original data set according to specifications. ● Download and investigate the Yolov8 model resources. ● Generate results from Yolov8 pretrained weights for detection (using included person class). ● Download the custom labeled data set (done for you, no need to hand label) and train a new Yolov8 model on these images. ● Use custom weights for person detection on the data set. ● Generate results column in metadata .csv of pretrained and custom weighted person detections. ● Complete gradient boost and linear regression analysis of appropriate variables (detected persons and popularity score) metadata.
Slide and Presentation	● <u>Goal</u>: Show what you know! Clearly explain the process you follow in reproduction and convey what you learned to your peers. ● <u>Slides</u>: ○ 7 slides for the presentation. ▪ Slide topics in order: Title Slide, Project Details, Data Acquisition, Modeling Approach, Results/Conclusions, What You Learned, References and Questions. ○ Slides submitted as PDF on Canvas. ● <u>Presentation</u>: ○ 5 minute duration. ○ Meant as an overview of your case process. ○ Focus on the process of the case, not so much the technical backing for the case.
Written Reflection	● <u>Goal</u>: Synthesize what you did and what you learned. ● Format: One to two pages. ○ Process of Reproduction - Two Paragraphs ○ What skills you practiced or learned - Two skills, one paragraph on each. ▪ Mention how each skill can apply to future data science projects. ▪ Can be quantitative skill or qualitative skill. ● Size 11 font. 1.15 spacing. Arial. ● PDF format submitted on Canvas.